

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m²

Renovation Details by Scenario

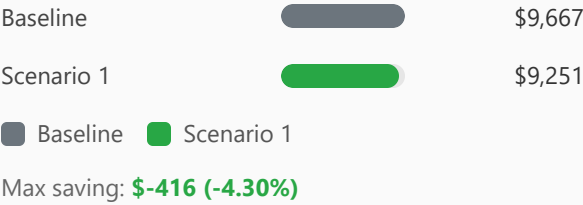
Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	unknown	Envelope renovation applied (details generated at report time)

Annual Energy Analysis

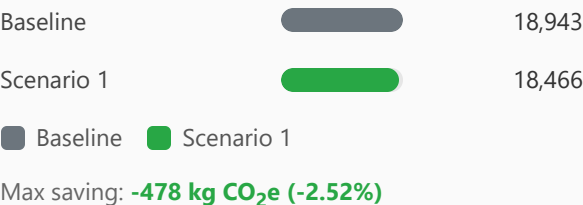
Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	195.48	190.41	5.07	2.59%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison



Min Retrofit Construction Cost

\$234,748

Scenario 1

Min Retrofit Embodied Carbon

18,714 kg CO₂e

Embodied carbon intensity (ECI): 37 kgCO₂e/m²
Scenario 1

Lowest Retrofit Cost Payback Period

564 years

Scenario 1

Lowest Retrofit Carbon Payback Period

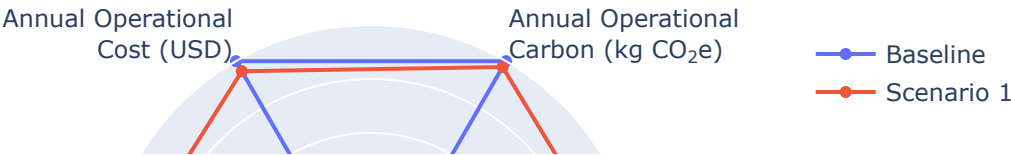
39 years

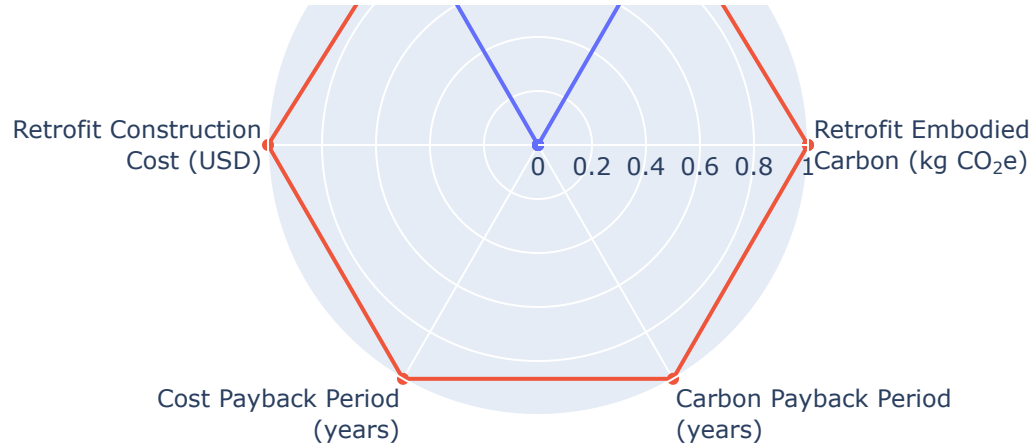
Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)





Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1 564 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1 39 yrs

Material List

No material consumption data found for renovation scenarios.

Result Summary

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	18,943	\$9,667
Scenario 1	18,714	\$234,748	18,466	\$9,251

Generated: April 27, 2026 | Report Type: Retrofit Impact Analysis | Run: run_test_005